श्रद्धावान लभते ज्ञानम्

Good Education, Good Jobs

Institute of Engineering and Management

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GROUP B

SUBJECT Advanced Programming(OOP) LAB

Personal Book Universe Tracker

1. Introduction

The *Personal Book Universe Tracker* is a Java-based application designed to organize and manage fictional universes, series, books and characters. The project aims to provide a structured way for users to explore relationships between different literary elements in a fictional multiverse.

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2. Objectives

- To implement Object-Oriented Programming (OOP) concepts in Java
- To design a hierarchical data structure for fictional universes
- To develop a simple GUI using Java Swing
- To allow users to view and manage books and characters efficiently

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3. System Design

3.1 Architecture

- **Model Layer** → Core entities
- **Management Layer** → Logic handling
- **Main Layer** → Initial point
- **GUI Layer** → User interaction

3.2 Class Structure

- Universe
- Series
- Book (Base Class)
- Novel (Subclass)
- FicCharacter

3.3 Relationships

- Universe → Series
- Series → Book
- Book → Character

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4. Object-Oriented Programming Concepts Used

✓ **Encapsulation**

Encapsulation is achieved by keeping class variables private and accessing them through public getter and setter methods, ensuring data security and controlled access.

✓ **Inheritance**

Inheritance is used where the Novel class extends the Book class, promoting code reusability and hierarchical structure.

✓ **Abstraction**

Abstraction is implemented by hiding internal implementation details and exposing only essential functionalities through the LibraryManager and GUI.

✓ **Polymorphism**

Polymorphism is demonstrated using base class references (Book) to handle subclass objects (Novel), allowing flexible and dynamic behaviour.

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5. Implementation Details

5.1 Technologies Used

- Java (Core + OOP)
- Java Swing (GUI)

5.2 Key Components

LibraryManager

- Manages all data and operations

DataInitializer

- Loads sample data

Main

- Contains the main() method

MainWindow (GUI)

- Handles user interaction

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6. Features

- Hierarchical organization
- GUI-based interaction
- Preloaded data
- Modular design

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7. Challenges Faced

- GUI layout issues
- Input handling errors
- Class relationship design
- Debugging compilation errors

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8. Limitations

- No database support
- Limited search functionality
- Basic UI

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9. Future Enhancements

- Database integration
- Improved UI

- Advanced search
- File storage

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10. Learning Outcomes

- OOP concepts (Encapsulation, Inheritance, etc.)
- Java Swing GUI development
- Debugging and error handling
- Modular system design

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11. Conclusion

The project successfully demonstrates the implementation of Object-Oriented Programming principles along with GUI development using Java Swing. It provides a structured and scalable approach to managing fictional universes and their elements. The system can be further enhanced with advanced features, making it a strong foundation for future development.